

—> **JavaScript Practice Theoretical Question -**

1. What is JavaScript?
2. What are the different data types in JavaScript?
3. What is the difference between null and undefined?
4. What is NaN?
5. Difference between let, const, and var.
6. What is type coercion?
7. Difference between == and ===.
8. What are truthy and falsy values?
9. What is the typeof operator?
10. What is a function?
11. Difference between function declaration and function expression.
12. What is an arrow function?
13. Difference between regular functions and arrow functions.
14. What is a callback function?
15. What is a higher-order function?
16. What is the purpose of the return keyword?
17. What are default parameters?
18. Difference between map(), filter(), and reduce().
19. What does forEach() return?
20. Difference between slice() and splice().
21. What is array destructuring?
22. What is an object?
23. What are Object.keys(), Object.values(), and Object.entries()?

24. Difference between `Object.seal()` and `Object.freeze()`.
25. What is object destructuring?
26. What is a class?
27. What is a constructor?
28. What is this keyword?
29. What is Object-Oriented Programming?
30. Explain encapsulation.
31. Explain inheritance.
32. Explain polymorphism.
33. What is synchronous programming?
34. What is asynchronous programming?
35. What is a Promise?
36. Explain Promise states.
37. Difference between `setTimeout()` and `setInterval()`.
38. What does `Promise.all()` do?
39. Difference between `Promise.race()`, `Promise.any()`, and `Promise.allSettled()`.
40. Difference between Promise and callback.

—> **JavaScript Practice Practical Question -**

1. Create variables for:
2. Name, Age, City. Print them in one sentence.
3. Find whether a number is even or odd.
4. Find the largest of three numbers.
5. Calculate simple interest.
6. Check whether a year is a leap year.
7. Create a grading system:

90+ = A

80+ = B

70+ = C

Below 70 = Fail

8. Check if a number is positive, negative, or zero.
9. Create a calculator using a switch.
10. Check whether a character is a vowel or consonant.
11. Print numbers from 1 to 100.
12. Print even numbers between 1 and 50.
13. Print multiplication table of 5.
14. Find the factorial of a number.
15. Check whether a number is palindrome.
16. Reverse a number using while.
17. Count digits in a number.
18. Create a function to add two numbers.
19. Create an arrow function that checks if a number is even.
20. Create a function with default parameters.
21. Create a function that returns the largest of two numbers.
22. Find the largest element in an array.
23. Count even numbers in an array.
24. Find the sum of all array elements.
25. Sort an array in ascending order.
26. Reverse an array.
27. Merge two arrays.
28. Find missing numbers from an array.
29. Multiply every element by 2 present in an array.
30. Get all even numbers from an array.

31. Convert all names to uppercase using `map()`.
32. Filter students with marks greater than 80.
33. Find average marks using `reduce()`.
34. Create a student object.
35. Add a new property to an object from the object.
36. Delete a property from the object.
37. Use object destructuring.
38. Use `Object.keys()`.
39. Use `Object.values()`.
40. Use `Object.entries()`.
41. Freeze an object and test it.
42. Create a Person class.
43. Create a Student class that inherits from Person.
44. Implement abstraction using a base class.
45. Create a Promise that resolves after 2 seconds.
46. Create a Promise that rejects after 3 seconds.
47. Convert a Promise example to `async/await`.
48. Use `Promise.all()` with three promises.
49. Use `Promise.race()` to the same.
50. Use `Promise.any()` to the same.
51. Use `Promise.allSettled()` to the same.
52. Print "Hello" every second and stop after 5 seconds.
53. Print multiplication table of 5 every second.
54. Stop `setInterval()` after 10 seconds.
55. Implement encapsulation using private fields.